What Could Policymakers Have Known? Employment Risk and Administrative Data in the Dutch NOW Programme

Abstract

During COVID, many Dutch businesses lost jobs, but some of those jobs would have been lost anyway, even without a crisis. This study asks: using only information the government already had at the time, can we estimate how many jobs each business would lose because of COVID, separately from the jobs it would have lost anyway?

We built two logistic regression models on about 10 million firm-year observations from the Dutch LISA employment register (2000–2016). One predicts whether a firm closes within three years, the other whether a firm cuts staff within one year. Each prediction is split into two parts: a structural part, based on pre-crisis firm characteristics (size, age, sector, province and workforce composition), and a crisis part, based on the revenue the firm lost.

The two parts could be separated. Structural risk follows sensible patterns: smaller, younger firms are most likely to close, while larger firms are more likely to cut staff but keep operating. Out-of-sample accuracy is moderate (AUC 0.69 for closure, 0.65 for staff cuts).

These are estimates, not measurements. The models learn from earlier, smaller crises, revenue loss is reconstructed rather than observed, and what would have happened without COVID cannot be checked. The predictions are reliable for groups of firms, not for individual firms.

Introduction

The COVID-19 pandemic prompted governments worldwide to deploy large-scale wage subsidy programmes at unprecedented speed. In the Netherlands, the Noodmaatregel Overbrugging voor Werkgelegenheid (NOW) provided wage compensation to firms reporting revenue losses, disbursing approximately €22 billion across eight rounds between 2020 and 2022 (UWV, 2022). Designed as a broad liquidity bridge, the programme prioritised rapid implementation over targeted support to firms most at risk of job destruction (Staatscourant, 2020).

Speed came at the cost of precision. NOW paid any firm reporting at least a 20 percent revenue loss, a threshold the Algemene Rekenkamer (2022) notes was chosen for rapid implementation rather than targeting accuracy. Broad eligibility meant support reached firms that would have contracted regardless of the pandemic alongside those whose employment losses were primarily crisis-induced. At the time, policymakers could not observe how many jobs fell into each category.

Whether this distinction could already have been drawn from information available during programme implementation is an empirical question with direct policy relevance. Emergency wage subsidies involve substantial public expenditure, and Rabobank (2024) estimates that the corona support measures prevented at least 11,000 bankruptcies from 2020 through 2027, but also sustained some non-viable firms too long. Telling apart job loss caused by the crisis from job loss that would have happened anyway is a first step towards understanding what such support can and cannot achieve.

This study therefore addresses one question: can we estimate the additional job loss caused by the COVID shock, separately from job loss that would have happened without it, using only information available at the time?

To answer it, we estimate two logistic regression models on administrative data of Dutch firms: one for firm exit and one for staff reductions at surviving firms. Each predicted probability is split into a structural component, based on pre-crisis firm characteristics, and a crisis-induced component, based on reported revenue loss. The contribution of this study is to show that this separation is feasible using real-time administrative data. The estimates are predictions that cannot be checked against what would have happened without COVID, so they are informative for groups of firms rather than for individual firms.

2. Literature Review

This question requires three kinds of background: why distinguishing structural weakness from crisis distress matters, what administrative data can predict about firm exit before outcomes are observed, and how employment loss unfolds, either through firm exit or through workforce reductions at surviving firms. This review covers each in turn and ends with the gap this study fills.

2.1 Structural weakness and crisis shocks as separate sources of risk

Wage subsidies respond to a specific market failure. In a sharp downturn, a viable firm can fail because a temporary drop in revenue leaves it unable to cover fixed labour costs (Gourinchas et al., 2021). When a firm closes, laid-off workers suffer lasting earnings and skill losses (Blanchard and Summers, 1986).

Because eligibility was broad, NOW reached two kinds of firm: some already structurally weak before the pandemic and others that were healthy beforehand but affected by the temporary shock. Caballero, Hoshi and Kashyap (2008) describe the first group as structurally weak firms operating in declining markets that would have contracted regardless of any crisis. Gourinchas et al. (2021) show that many firms affected by COVID-19 were viable before the pandemic and faltered only because demand collapsed. The literature therefore treats structural weakness and crisis shocks as distinct sources of exit risk. A broad subsidy reaches both without distinguishing between them, so the share of job loss caused by the crisis rather than by pre-existing weakness cannot be observed directly.

The distinction also matters for longer-run outcomes. Cheap credit in Japan kept unproductive firms alive and diverted resources from healthier competitors (Caballero, Hoshi and Kashyap, 2008), while González, Moral-Benito and Soler (2023) show that closures raise productivity at normal exit rates but lower it at very high ones.

2.2 Predicting firm exit from administrative data

Firm exit can be predicted from observable firm characteristics. Altman (1968) showed that a small number of financial ratios could identify failing firms years in advance. Shumway (2001) improved on this by following firms over time, showing that firm age and size matter alongside financial variables. Lessmann et al. (2015) demonstrate that more complex machine-learning methods often improve predictive accuracy but at the cost of interpretability.

The most accurate models typically rely on detailed financial information. Davies, Kattenberg and Vogt (2023) combine machine learning with administrative records and confidential balance-sheet information to predict Dutch firm exits substantially better than standard approaches. They also show that much of this performance can be retained with a considerably smaller set of predictors, and that financially weaker firms were less likely to apply for NOW, so recipients are not representative of all firms.

Administrative data therefore contain useful information about firm vulnerability, available before employment outcomes are observed. Existing studies use it to predict firm exits, but they do not separate structural weakness from crisis exposure.

2.3 How employment loss unfolds during a crisis

Employment loss occurs through two distinct channels. Firms may reduce employment while continuing to operate, or they may exit the market altogether. Davis, Haltiwanger and Schuh (1996) show that both channels contribute substantially to gross job destruction and must be modelled separately to capture aggregate employment dynamics. A framework for predicting crisis-induced employment loss must therefore account for both channels.

2.4 The gap this study fills

The literature provides each element separately: structural weakness and crisis exposure are recognised as distinct sources of risk, administrative data contain predictive information about firm vulnerability, and employment loss unfolds through both firm exit and workforce shrinkage. What has not been done is combining these elements. No existing study separates predicted structural from crisis-induced employment loss at the firm level using only real-time administrative information.

This study develops such a separation and applies it to Round 1 NOW recipients, using only information available during programme implementation. Whether such a separation is feasible in real time is the question the remainder of the paper addresses.

3. Methodology

3.1 Overview

The literature establishes that firm exit can be predicted from administrative data (Section 2.2), that structural weakness and crisis exposure are distinct sources of risk (Section 2.1), and that employment loss unfolds through both firm exit and layoffs at surviving firms (Section 2.3). This section develops a framework that combines these elements to answer the paper's question.

The framework uses two logistic regression models estimated on historical LISA data. The exit model predicts the probability that a firm disappears from the register over a three-year horizon. The layoff model predicts the probability that a surviving firm reduces headcount within one year. Each model's total predicted probability decomposes into a structural component, driven by pre-crisis firm characteristics, and a crisis-induced component, driven by reported revenue loss and a common crisis indicator. Applied to Round 1 NOW recipients, this decomposition provides the empirical answer to the research question.

The framework does not evaluate whether NOW worked. It asks what predicted employment risk looked like at the time of programme implementation.

3.2 Model Specification

The research question rests on two prediction models, specified here. Employment risk is estimated from the LISA employment register, covering 2000 to 2016, using two logistic regression models that share the same specification but differ in their dependent variable and horizon.

The two models reflect the two channels of employment loss. Firms may reduce employment through layoffs while continuing to operate, or they may exit the market altogether. These channels operate on different timescales. Layoff decisions respond to revenue shocks within months, so the layoff model uses a one-year horizon; extending it would blur the immediate crisis response with subsequent unrelated shocks. Firm exit responds more slowly because firms exhaust reserves, defer payments, and restructure before failing, so the exit model uses a three-year horizon; a shorter window would systematically underestimate crisis-induced exits that had not yet materialised.

Firm exit here captures the disappearance of the firm from administrative records, regardless of the specific legal pathway. The layoff model treats any decrease in headcount within a year as a layoff event. Because the framework is predictive, this definition does not need to isolate involuntary separations from natural attrition: structural features capture each firm's typical rate of headcount reduction from the training years, so the baseline already reflects normal turnover. The crisis component then identifies the additional layoff probability associated with crisis conditions, on top of that baseline.

Both models share the following specification. For firm i in sector s at time t:

$$\text{Logit}(P(Y_{i□□} = 1)) = \alpha + \beta_1 X_i + \beta_2 TWV_{□□} + \beta_3 D_{□} + \beta_4 (D_{□} \times X_i) + \gamma_{□} + \delta_{□} + \varepsilon_{i□□}$$

where X_i represents firm characteristics, TWV□□ is a sector-level shock proxy (defined below), D□ is a crisis dummy, and γ□ and δ□ are sector and province fixed effects. In the exit model, Y_i□□ equals one if the firm closes within three years; in the layoff model, Y_i□□ equals one if a surviving firm loses at least one employee within one year. The exit model omits the crisis dummy and its interactions because they are collinear with reported revenue loss at deployment (Appendix G).

The crisis component combines two distinct deviation measures. The crisis dummy captures years of generally elevated exit and layoff risk across the economy. TWV captures how much a given sector's value added falls short of that sector's own long-run average trend in a given year, independent of whether the year is difficult for the whole economy. Structural features capture the firm-level baseline against which both deviations are measured, so that the crisis dummy and TWV identify deviations from each firm's own baseline rather than differences between firms.

The structural predictors X_i are pre-crisis firm characteristics recorded in administrative data available for all firms: firm size (log employees), firm age (four bands), two-digit sector, province, and workforce composition (full-time share, female share, and female full-time share). Because the framework is predictive rather than causal, structural features are included as fully as the administrative data allows, so that the crisis component more accurately reflects crisis-induced risk rather than unmodelled structural variation. Age enters in bands because the training data tops out at roughly 16 years, while FIRMBACKBONE at deployment contains substantially older firms. The estimation sample is restricted to firms with at least two employees (Appendix D).

Logistic regression is chosen because the framework requires extrapolation beyond the historical training range. COVID-19 revenue losses are substantially larger than the historical shocks used for estimation, and tree-based models cannot extrapolate to feature combinations outside those seen in training. A logit estimates a continuous relationship between shock size and outcome probability that remains defined for out-of-sample shock magnitudes. A hazard model would allow more exact timing of firm exits but is not needed here, since the objective is to decompose expected employment loss over fixed prediction horizons rather than to predict the timing of individual events.

Revenue loss is not observed in the training data, since firms only reported it once NOW existed. The historical models therefore train on TWV, measured at the parent-sector letter level and demeaned by the sector's long-run growth so that it reflects below-trend shocks (Appendix A). The years 2009, 2010, 2012, and 2013 are marked as crisis years. The crisis dummy is interacted with sector, firm size, firm age, and full-time share, because industries face different demand shocks, smaller firms have less financial slack, young firms lack retained earnings, and firms with higher fixed labour costs face sharper adjustment pressures. A single TWV coefficient is estimated without interactions, avoiding overfitting on the limited number of crisis observations (Appendix B).

During deployment, the crisis marker is set to one for every NOW recipient and the sector-level TWV proxy is replaced by each firm's reported quarterly revenue loss, multiplied by the coefficient estimated for TWV. Since revenue loss is not directly observed, it is reconstructed from NOW advance payments (Appendix C). Reported revenue loss enters as

a raw percentage, whereas TWV is a deviation from sector-average growth, which introduces a small level inconsistency. Because Dutch sector growth is typically 1–3 percent annually and COVID revenue losses range from 20 to nearly 100 percent, the distortion is small (Appendix A).

3.3 Expected Job Loss

This section converts the predicted probabilities from Section 3.2 into firm-level expected job loss.

For the exit model, the predicted probability decomposes as $p = p_0 + \Delta p$, where p_0 is the structural probability of exit and Δp the additional probability attributable to the crisis. Expected employment loss from exit equals the predicted exit probability multiplied by firm employment. Because the model output is a probability, these are probabilistic quantities rather than forecasts for individual firms, and they are interpretable only in aggregate.

Because the exit model predicts cumulative probability over three years, the cumulative probability is assumed to increase linearly across the three years (Appendix D), giving an annual survival probability for each year. This is used when calculating expected losses from layoffs, since a firm already expected to have exited cannot also reduce its workforce further.

For the layoff model, the crisis shock is assumed to affect only the first year, so the first-year layoff probability is $q_1 = q_0 + \Delta q$, while the second and third years use the structural probability q_0 only. Expected employment loss from layoffs is $(1 - p_{\square}) \times q \times L$, where $1 - p_{\square}$ is the probability the firm remains in operation in year t, q is the annual layoff probability, and L is the average number of jobs lost conditional on a layoff event, estimated separately by firm size (Appendix F). Weighting by the probability the firm still operates prevents the two channels from together exceeding a firm's employment.

4. Data

4.1 LISA Employment Register

The primary data source is the LISA employment register (LISA, n.d.), a Dutch administrative dataset covering virtually all firms operating in the Netherlands. LISA records firm-level employment, sector classification, province, size category, and year of founding for each firm annually. The training panel covers 2000 to 2016, yielding approximately 10 million firm-year observations after restricting to firms with at least two employees. Sectors A

(Agriculture, Forestry and Fishing) and B (Mining and Quarrying) are not included in the GOLD training extract and therefore do not appear in the estimation sample. However, these sectors account for only a small share of Round 1 NOW recipients (Smits et al., 2022), suggesting that their exclusion is unlikely to materially affect the results. The 2019 LISA wave provides pre-crisis firm characteristics for the deployment sample. Both are accessed through the FIRMBACKBONE infrastructure (FIRMBACKBONE, n.d.), which links LISA observations across years using a consistent firm identifier.

4.2 NOW Administrative Data

NOW data are originally provided by UWV (UWV, n.d.) and cover all eight rounds of the programme. The data record the advance payment received by each firm in each round, from which each firm's revenue loss is reconstructed (Appendix C). Through the FIRMBACKBONE infrastructure (FIRMBACKBONE, n.d.), these records are assigned a consistent firm identifier that enables linkage to the LISA employment register, using the 2019 LISA wave where available and subsequently matching remaining firms to the 2020, 2021 and 2022 LISA waves.

4.3 Data Linkage and Sample Construction

NOW recipients are linked to the LISA employment register via the FIRMBACKBONE identifier, first to the 2019 wave and then, where unmatched, to the 2020, 2021, and 2022 waves. Of 81,339 Round 1 recipients, 51,044 (62.8 percent) are matched. Matched firms receive larger average subsidies than unmatched ones (€65,719 versus €51,477), indicating a mild size bias in linkability. After excluding firms with missing employment or implausible founding years, the final deployment sample contains 194,161 firm-round observations across the eight rounds. Because unmatched firms are smaller and reconstruction error is largest among the smallest firms (Appendix C), the sampling bias and the measurement error run in the same direction: the firms that drop out are also the ones measured least reliably. The analytical sample is therefore better measured than the full population, at the cost of underrepresenting the smallest firms.

Sector	N firms	Median jobs	Median age	Mean rev. loss <8 (%)	Mean rev. loss >8 (%)	Mean rev. loss (%)	Total NOW budget (€M)	% of total budget
G	13,809	6.0	3	49.3%	44.2%	48.1%	848.7	20.8%
C	4,625	11.0	2	38.0%	33.5%	37.2%	622.0	15.3%
N	3,228	5.5	2	65.7%	58.4%	64.5%	468.2	11.5%
J	2,116	6.0	1	43.6%	38.2%	42.7%	431.3	10.6%
I	8,226	8.0	2	70.4%	60.7%	68.8%	430.1	10.6%
M	5,288	6.0	2	44.4%	37.7%	43.1%	375.0	9.2%
H	1,985	9.0	1	42.5%	34.5%	41.2%	325.6	8.0%

R	2,083	7.0	2	61.0%	54.4%	60.1%	161.9	4.0%
F	2,130	9.0	2	37.1%	29.9%	35.6%	118.4	2.9%
Q	2,820	6.0	2	42.6%	34.3%	41.4%	102.6	2.5%
P	983	5.0	2	45.4%	40.9%	44.5%	49.3	1.2%
S	1,662	4.0	3	58.8%	48.3%	56.5%	46.4	1.4%
K	372	6.0	2	33.5%	26.6%	32.2%	27.2	0.7%
L	635	5.0	3	37.5%	33.4%	36.9%	19.9	0.5%
E	110	9.0	2	34.8%	37.3%	35.5%	6.5	0.2%
D	13	5.0	2	41.7%	15.3%	37.6%	0.9	0.0%
O	6	108.0	2	8.7%	0.0%	7.3%	0.4	0.0%
Total / Avg	**50,091**	**12.7**	**2.1**	**44.4%**	**36.9%**	**43.1%**	**4034.4**	**100%**

Table 4.1. Round 1 NOW applicants by sector. N firms is the number of firms in the matched analytical sample. Mean rev. loss is the average implied revenue loss reconstructed from NOW advance payments. Total NOW budget is total round 1 expenditure in the matched sample in millions of euros. % of total budget is each sector's share of total round 1 matched expenditure (€4.07 billion). Sectors are sorted by total NOW budget. The nine largest market sectors (C, F, G, H, I, M, N, R, S) together account for the large majority of firms and expenditure and form the estimation sample for the crisis component of the exit and employment loss models. Excludes sector A and B, and firms with missing values.

Two features of Table 4.1 matter for what follows. First, expenditure is highly concentrated: wholesale and retail (G), manufacturing (C), and administrative services (N) together account for roughly half of the €4.07 billion in matched Round 1 spending. Second, reported revenue losses vary widely across sectors, from around 36 percent in construction (F) to nearly 69 percent in hospitality (I), foreshadowing the crisis-exposure differences that drive the later results. The crisis component of both models is estimated on the nine largest market sectors only(Appendix B).

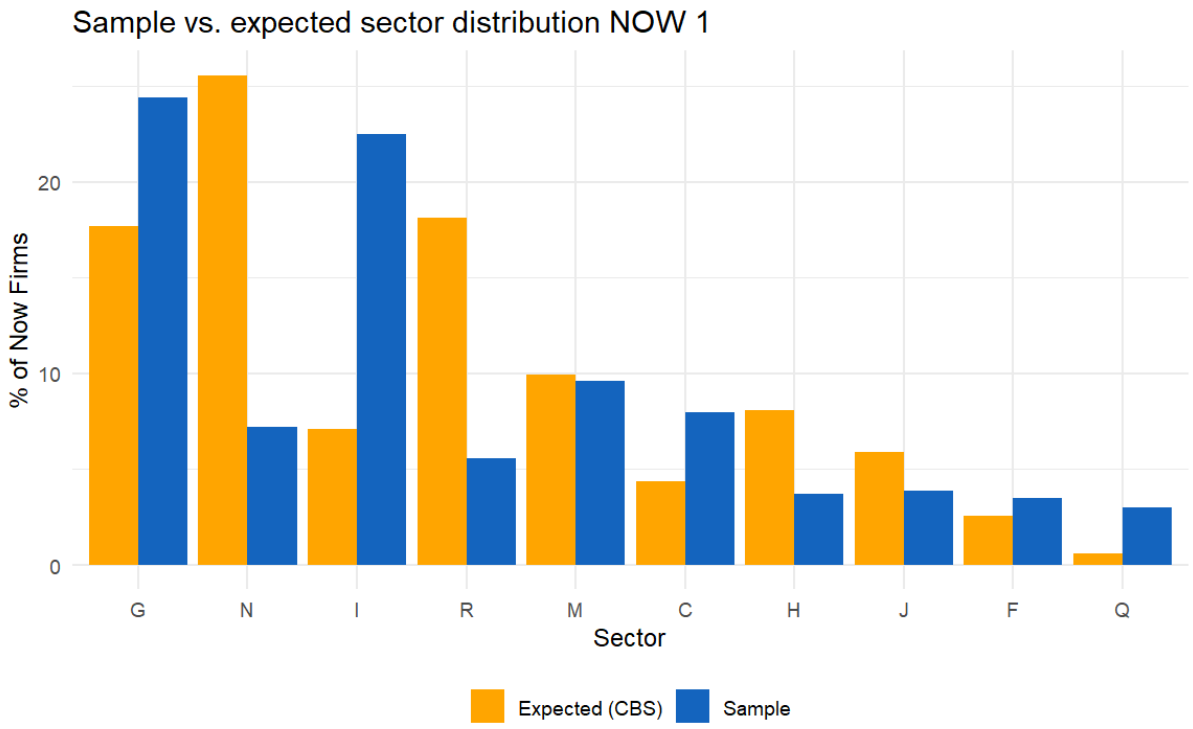


***Figure 4.2**. Matched round 1 sample versus expected NOW round 1 sector distribution (CBS estimates); the sample broadly tracks the population, with gaps concentrated in small-firm sectors.*

Figure 4.2 compares the sector distribution of the matched sample against the expected distribution of NOW round 1 recipients implied by the CBS participation estimates in Smits et al. (2022). These gaps likely reflect differences in how consistently firms can be linked to the LISA register, and the resulting sector imbalances should be kept in mind when interpreting sector-level results.

4.4 Supplementary Data

Sector-level value added growth (TWV) is obtained from CBS StatLine (CBS, n.d.-a) at the parent sector letter level for 2000 to 2020, and is used as the crisis shock measure in model training. Sector average wages including employer social contributions are obtained from CBS National Accounts (CBS, n.d.-b) at the two-digit SBI level and are used to reconstruct revenue loss from NOW advance payments (Appendix C). Firm exit by 2022 is determined from the 2022 wave of FIRMBACKBONE (FIRMBACKBONE, 2022), which incorporates the LISA employment register (LISA, 2016) used for the 2000–2016 training data. A wage data overview is given in Appendix J.

5. Results

5.1 Structural and Crisis Components of Risk

This section answers the research question: whether the additional job loss caused by the COVID shock can be estimated separately from job loss that would have happened regardless.

Both models achieve moderate out-of-sample discrimination: AUC = .69 for the exit model, AUC = .65 for the layoff model. This confirms that pre-crisis firm characteristics alone predict which firms are structurally more or less likely to exit or lay off workers, independent of any crisis. The structural components of the two models point in different directions, consistent with the exit-versus-layoff distinction motivated in Section 2.3. In the exit model, a doubling of firm size is associated with roughly a 32 percent reduction in the odds of exit ($\beta = -0.458$), and firms aged 0–3 years face substantially higher exit risk than firms aged 16 or older ($\beta = 0.667$): smaller, younger firms are structurally most exit-prone. In the layoff model, a doubling of firm size is associated with roughly a 35 percent increase in the odds of a layoff ($\beta = 0.437$), and a higher full-time share reduces layoff risk ($\beta = -0.301$): larger firms with more flexible workforce composition are more likely to adjust employment while continuing to operate rather than exiting.

The crisis component is the additional probability of exit or layoff attributable to the COVID shock, captured through both the crisis dummy and reported revenue loss (Section 3.2). The two channels contribute differently across the two models. The exit model omits the crisis dummy because it is collinear with reported revenue loss at deployment (Appendix G), so the entire crisis component in the exit model is captured by revenue loss alone ($\beta = -0.016$ per percentage point of reported loss). The layoff model retains both: a crisis dummy capturing the additional layoff probability associated with an economy-wide downturn, independent of reported revenue loss ($\beta = -0.053$), and revenue loss capturing the additional layoff probability specific to the firm's own shock ($\beta = -0.014$ per percentage point). The crisis dummy's interactions shift this economy-wide effect towards older, larger, and more full-time-intensive firms, while its overall contribution remains small relative to revenue loss.

Together, these results show that structural and crisis-induced risk are separately identifiable from pre-crisis characteristics and reported revenue loss. The two models discriminate reasonably well out of sample, their structural components reflect economically sensible firm-level patterns, and their crisis components respond specifically to the crisis rather than duplicating the structural signal. Full coefficients, standard errors, and specification stability checks for both models are reported in Appendices G and H.

5.2 Expected Job Loss

Applied to the matched sample of 50,091 Round 1 recipients, which together employ roughly 940,000 people, the models predict roughly 250,000 jobs lost over a three-year horizon. Of these, roughly 225,000 (24 percent of employment) are structural and roughly 27,000 (3 percent of employment) are crisis-induced. The crisis-induced component therefore accounts for roughly 11 percent of predicted job loss.

The crisis-induced figure reflects the expected effect of a roughly three-month crisis (Section 3.3), so it is a conservative estimate of the full crisis impact. These are aggregate expectations rather than forecasts for individual firms: firm-level predictions carry substantial

noise from reconstructed revenue loss and unobserved firm characteristics, and are informative only when summed across many firms.

6. Discussion

6.1 What This Study Establishes

A government designing a wage subsidy under crisis conditions faces a foundational problem: knowing who is hit. To concentrate support on firms whose employment is truly at risk because of the crisis, it needs to estimate expected COVID-19 induced job loss above what would have occurred under normal economic conditions. This study asks whether that separation of structural and crisis-induced employment risk is possible using only real-time information.

Section 5 shows that it is, within limits. Two logistic regression models, estimated entirely on pre-crisis data, separate structural from crisis-induced employment risk. Their structural components reflect economically sensible patterns, and their crisis components respond specifically to reported revenue loss rather than duplicating the structural signal. Applied to Round 1 NOW recipients, the models predict that roughly 11 percent of expected job loss was crisis-induced, with the rest structural. Because the crisis-induced component is scaled to a roughly three-month crisis, this is a conservative estimate of the full crisis impact.

6.2 What Could Have Been Known in Real Time

Every input to the analysis (pre-crisis firm characteristics, the historical relationship between economic shocks and employment adjustment, reported revenue loss, and sector wages) was available to the government before Round 1 was disbursed. The separation could therefore in principle have been made while the programme was running, rather than only in hindsight.

6.3 Limitations

- **Extrapolation of the crisis coefficient:** The crisis component is estimated from historical downturns, primarily 2009–2013, and applied to COVID-19. Historical shocks were substantially smaller than COVID-19 revenue losses, and the framework assumes the estimated relationship between shock size and employment adjustment extends linearly to much larger shocks.
- **Directional bias in reconstruction error:** Revenue loss is not directly observed and is reconstructed from NOW advance payments using sector-average wages. Larger firms disproportionately fell below the 20% eligibility threshold, suggesting their predicted crisis severity is underestimated. Conversely, the smallest firms disproportionately hit the 100% cap, suggesting their predicted crisis severity may be overestimated. This systematic size bias means the true difference in crisis-induced risk between large and small firms may be smaller than the model predicts.

- **No validation against what would have happened:** The predictions describe expected employment loss without COVID-19 and cannot be compared with an observed counterfactual, so they cannot be checked against actual outcomes.
- **Firm-level predictions are not decision-grade:** Individual firm-level predictions carry substantial noise from unobserved characteristics such as liquidity, management quality, and product niche. All findings in this paper are aggregate claims across groups of firms.
- **Revenue loss is not itself decomposed:** The framework treats reported revenue loss as a single input; it does not mathematically separate revenue loss into the shares attributable to the crisis versus pre-existing structural weakness.

6.4 Future Research

Two extensions would strengthen the framework. Incorporating firm-level financial information such as liquidity or leverage, where available, would allow the crisis response to vary across firms rather than assuming a common relationship with reported revenue loss. Applying the framework to a different crisis, where untreated firm outcomes are eventually observable, would allow the predictions made here to be validated against what would have happened without intervention.

Conclusion

This study develops a firm-level crisis risk framework and applies it to recipients of the Dutch NOW wage subsidy programme. By separating structural vulnerability from COVID-19 induced job loss using only real-time administrative information, it shows that the extra job loss caused by the crisis can be estimated before outcomes are observed. Two logistic regression models discriminate moderately well out of sample (AUC .69 for exit, .65 for layoffs), their structural components reflect sensible firm-level patterns, and their crisis components respond specifically to reported revenue loss. Roughly 11 percent of predicted job loss among Round 1 recipients is crisis-induced.

These estimates carry real limits: they extrapolate from smaller historical downturns, rely on reconstructed revenue loss, cannot be validated against a counterfactual, and are informative for groups of firms rather than individual firms. Within those limits, the separation provides the baseline that any assessment of crisis support needs: knowing who is hit by the crisis, and by how much.

Appendix A - Turnover Change as a Proxy for Value Added value

The exit and employment models are trained on sector-level value added change (TWV) as the crisis shock measure. Turnover measures a firm's total sales, whereas value added measures the value created after subtracting the cost of intermediate inputs:

$$Turnover = Sales$$

$$Value\ added = Sales - Intermediate\ inputs$$

Both measures therefore reflect changes in underlying economic activity, although value added excludes fluctuations driven purely by changes in purchased inputs. Because consistent historical value-added data are available across sectors and years, TWV is used to estimate the historical crisis models. At the time of intervention, however, value added is not yet observable. Policymakers instead observe firms' declared quarterly revenue losses through NOW applications.

This appendix examines whether sector-level turnover change provides a reasonable proxy for sector-level value added change, thereby assessing the validity of applying model coefficients estimated using TWV to firm-declared revenue losses.

To examine the relationship between the two measures, I regress 2020 sector-level value added change on sector-level turnover change across 51 two-digit SBI sectors using CBS StatLine data.

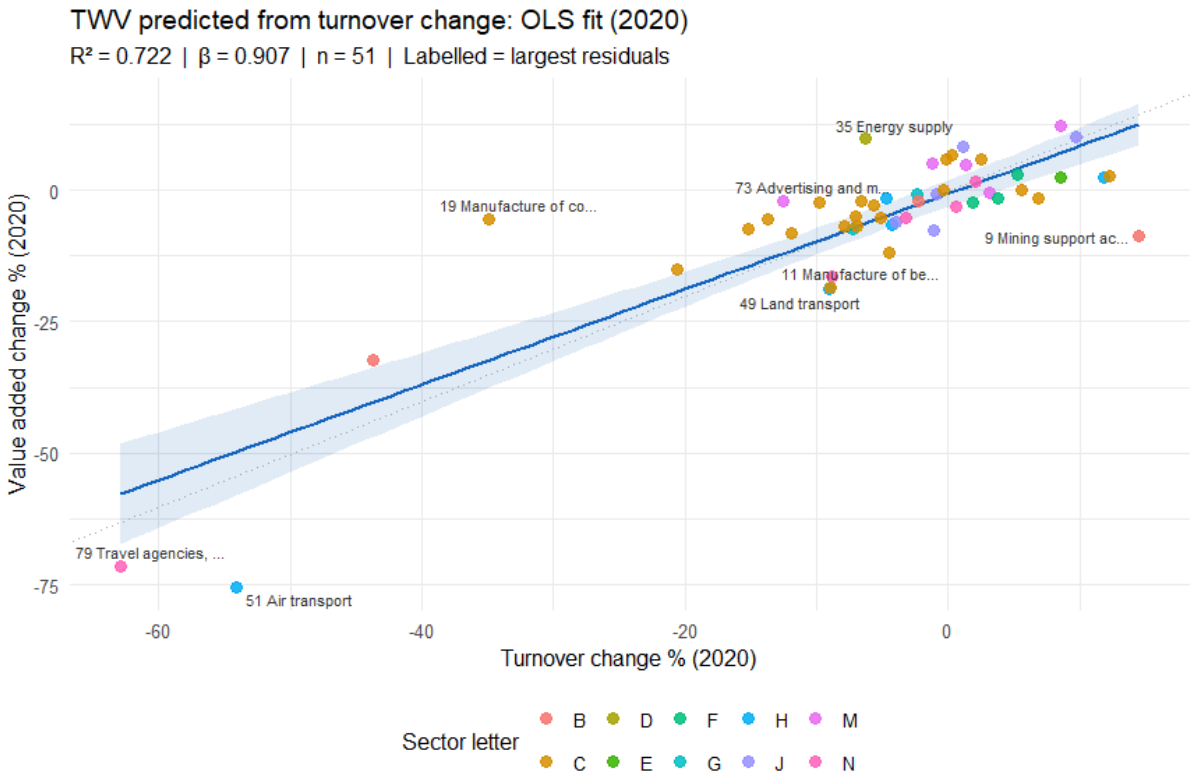


Figure A1*. Sector-level value added change (TWV) against turnover change in 2020 across 51 two-digit SBI sectors, with the OLS fit (R² = 0.722, β = 0.907); the two move closely together, with the largest residuals labelled.*

Regression analysis

The pooled OLS regression yields R² = 0.722 with β = 0.907, indicating that a one percentage point decline in turnover is associated with approximately a 0.907 percentage point decline in value added. The two measures move proportionally for most sectors, with the intercept of -0.694 being negligible in practice.

Adding sector letter fixed effects raises R² to 0.808, suggesting that broad sector categories account for some remaining systematic differences in the turnover-TWV relationship - likely reflecting structural differences in input intensity and cost rigidity across sector groups.

Within-sector patterns

The relationship is strongest and most linear in the sectors most affected by COVID-19 and most heavily represented in NOW applications transport and storage (H), temporary employment agencies and travel (N), and wholesale and retail trade (G)(Smits 2022). Sectors showing weaker within-sector fit, such as mining, utilities, and construction, represent a small share of NOW recipients and are characterised by fundamentally different revenue recognition and cost structures. The overall pattern suggests TWV and Revenue loss are similar for most sectors.

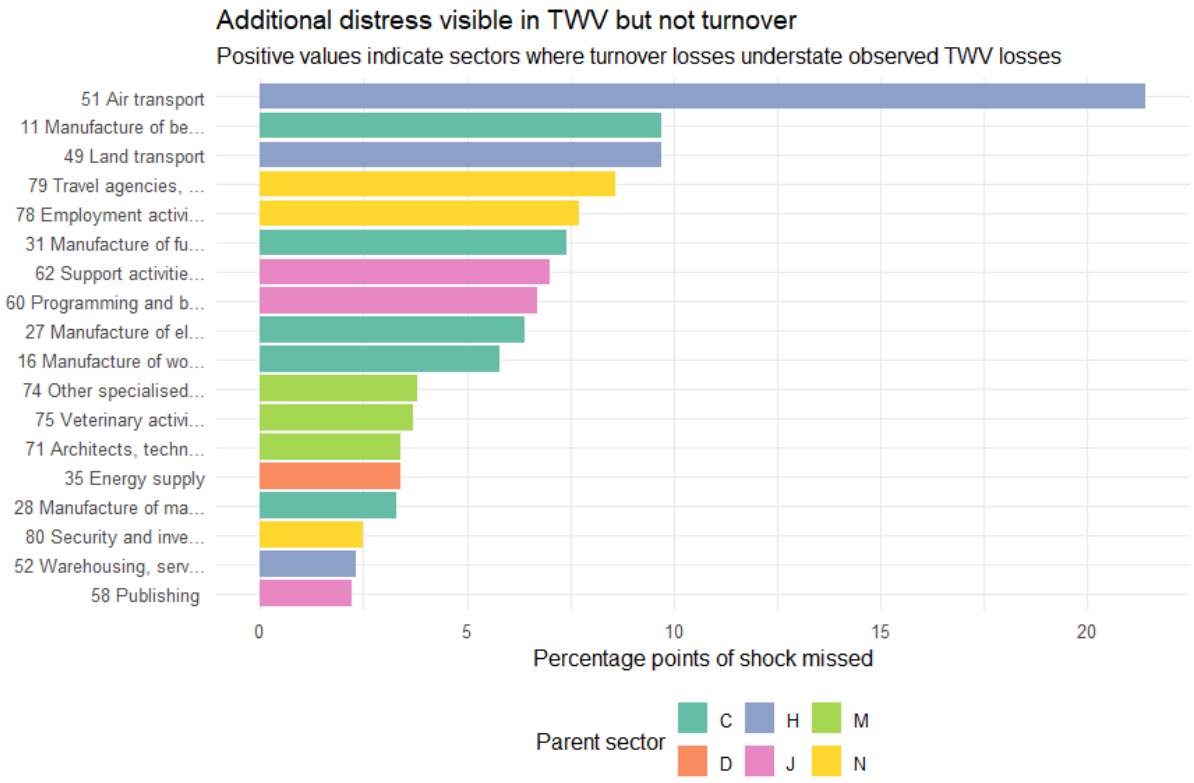


***Figure A2**. Sectors where turnover change understates the TWV shock, measured as the gap in percentage points between observed TWV decline and turnover decline in 2020; air transport shows the largest discrepancy at roughly 21 points, with most other sectors below 10.*

Where does it diverge?

Although turnover change and value added change are strongly correlated overall, Figure A2 shows several sectors where turnover losses understate observed TWV losses. When firm-reported revenue losses are substituted for TWV during deployment, these sectors therefore receive a weaker estimated crisis shock than implied by the underlying change in value added. This may cause genuinely distressed firms to be assigned lower predicted crisis risk.

The divergence is not random but appears to reflect differences in sector cost structures. Because value added equals turnover minus intermediate inputs, the two measures move together when input costs fall alongside sales. Where a larger share of costs remains fixed in the short run, value added declines more rapidly than turnover, causing turnover losses to understate the severity of the shock.

Figure A2 illustrates this pattern. The largest discrepancy occurs in air transport (SBI 51), where turnover declined by 54.1% while value added declined by 75.5%. Smaller but similar differences appear in transport, travel agencies, temporary employment services, and several manufacturing subsectors. These sectors combine relatively high fixed costs with strong exposure to COVID-19 demand shocks.

The divergence is concentrated in a limited number of sectors rather than being widespread across the economy. For most sectors, turnover and value added move closely together,

consistent with the strong relationship observed in Figure A1. The results therefore suggest that turnover is a reasonable deployment variable overall, while highlighting sectors where crisis severity may be understated.

Extension

The present analysis uses a single cross-section of 2020 sector-level data. Future work could strengthen this comparison by constructing a panel using multiple years of quarterly CBS observations, providing greater variation for estimating the relationship between turnover and value added across sectors and over time.

Appendix B: Model Specification - Interaction Term Justification

The models include interaction terms only where there is a clear theoretical reason to expect that a firm characteristic modifies the effect of an economy-wide crisis. This appendix explains the rationale for the selected interactions and why additional interactions were not included.

Both the exit and employment-loss models include interaction terms only where there is a plausible reason to expect that a firm characteristic modifies the effect of a crisis on employment risk. Limiting interactions to a small set of theoretically motivated variables keeps the models simple, reduces the risk of overfitting, and is intended to improve their ability to generalise to the much larger COVID-19 revenue losses observed at deployment.

Firm size interacts with the crisis dummy because larger firms may be better able to absorb temporary revenue losses through greater financial reserves, better access to credit, or more diversified sources of income, suggesting that the effect of a crisis may differ by firm size.

Firm age interacts with the crisis dummy because older firms may be more resilient to adverse shocks, having accumulated experience, established customer relationships, or financial buffers that could influence how they respond to a crisis.

Full-time share interacts with the crisis dummy because firms with a larger share of permanent employees may have less flexibility to adjust labour costs during a downturn, potentially altering the relationship between a crisis shock and employment loss.

Sector interacts with the crisis dummy because the economic impact of a recession is expected to differ across industries. Interactions are estimated at the one-digit sector level rather than the two-digit SBI level to ensure sufficient observations within each sector.

Female workforce share is not interacted with the crisis dummy because I do not identify a clear theoretical reason why it would systematically modify firms' sensitivity to economic shocks beyond the structural effects already captured in the model.

Revenue loss interactions

Revenue loss enters both models only as a continuous main effect. No interaction terms are included because COVID-19 revenue losses are much larger than those observed during the historical crises used for estimation, making interaction effects difficult to estimate reliably. In addition, firm characteristics such as size and age may already be partly reflected in observed revenue losses, so further interactions may capture the same information twice rather than improving prediction.

Sector	Corr. (threshold −5%)	Obs. (−5%)	Corr. (threshold 0%)	Obs. (0%)
C	-0.300	1	-0.343	7
F	-0.507	4	-0.526	7
G	-0.117	1	-0.221	7
H	-0.463	1	-0.393	8
I	-0.283	3	-0.510	8
M	-	0	-0.069	7
N	-0.360	4	-0.380	7
R	+0.187	1	-0.225	9
S	-	0	+0.288	9
Shock coef.	**−0.00186 (p = 0.001)**		**−0.00197 (p < 0.001)**	

Table B1. *Robustness of sector-level shock coefficient to crisis threshold choice. Corr. reports the within-sector correlation between the demeaned TWV shock and detrended exit rate. Obs. counts sector-year observations qualifying as crisis years under each threshold (−5%: shock below −5%; 0%: shock below 0%). Sectors M and S have no qualifying observations at the −5% threshold (shown as -). The pooled shock coefficient is stable across thresholds, changing from −0.00186 (p = 0.001) to −0.00197 (p < 0.001). The −5% threshold is retained in the final model. Only sectors included in the regression (C, F, G, H, I, M, N, R, S) are shown.*

Table B1 evaluates whether the estimated crisis effect depends on the choice of crisis threshold used to identify downturn years. Relaxing the threshold from −5% to 0% changes the estimated shock coefficient only from −0.00186 to −0.00197, indicating that the relationship between sector-level shocks and firm exits is largely unaffected by the threshold choice. The −5% threshold is therefore retained in the final specification.

Appendix C Revenue loss reconstruction

Firm level revenue loss is not directly observed in the dataset, to reconstruct it is derived from the NOW advance cost of each firm. The now advance cost of each firm is: NOW_cost = revenue_loss × wage_bill x 3 months × subsidy_rate x 0.8.(UWV).

where the subsidy rate is 0.9, the factor 0.8 represents that 80% of the cost is paid upfront, and the wage bill is approximated using sector wages, number of jobs and the duration of 1 round being 3 months.

Rearranging the formula gives: revenue_loss = NOW_cost / (sector_wage x 3 months × subsidy_rate x 0.8).

NOW Round	Number of Firms	Mean Rev. Loss (%)	Median Rev. Loss (%)	% at Cap (100%)
1	51,044	50.0	41.1	20.3
2	26,365	50.7	40.8	21.5
3	29,974	42.8	30.3	16.7
4	29,529	47.3	36.4	19.7
5	18,186	45.4	33.7	18.9
6	11,286	38.0	25.7	13.4
7	15,586	36.8	24.1	13.3
8	12,170	45.7	34.1	19.0

Table C1. *Revenue loss distribution by NOW round. Sample restricted to firms matched to LISA employment register. Cap at 100% refers to firms reporting complete revenue loss.*

Graph C1 shows the mean implied revenue loss is 50% for round 1, consistent with the UWV reported numbers of round 1 and 2. Values exceeding 100% are capped, as declared revenue loss cannot exceed 100% by definition.

Size Category	% at Cap (100%)	% Below 20%	Avg. NOW Budget (EUR)	Number of Firms
1	52.0	14.0	32,010	7,569
2	46.4	13.3	22,201	23,347
3	18.8	32.2	25,661	56,879
4	11.7	32.4	33,584	47,796
5	9.6	29.6	53,351	30,625
6	8.2	29.9	107,628	19,807
7	7.1	35.3	247,064	5,247
8	6.7	39.0	436,533	1,927

9	10.5	38.1	1,162,909	734
10	9.6	39.4	15,171,576	104
11	5.6	22.2	2,807,043	18
12	4.6	56.3	6,310,065	87

Table C2. *Revenue loss reconstruction quality and average NOW budget by CBS establishment size category. The share of firms at the 100% cap is strongly concentrated in the two smallest size categories, while the share below the 20% eligibility threshold is relatively higher for larger firms. Both patterns are consistent with reconstruction error arising from the use of sector-average wages as a proxy for firm-level wage bills. Average NOW budget increases substantially with firm size, reaching several million euros for the largest categories. Categories 10-12 contain few observations and should be interpreted with caution.*

Figure C2 illustrates the distribution of reconstructed revenue losses by firm size. Approximately 20% of firms reach the 100% cap, while around 29% fall below the 20% NOW eligibility threshold. Since all firms in the sample received NOW support, observations below 20% reflect reconstruction error rather than true ineligibility. The error arises because sector-average wages are only an approximation of firm-specific wage bills, causing estimated revenue losses to be either over- or understated. The concentration of capped observations among the smallest firms suggests that this approximation error is greatest where small differences in wages translate into large percentage differences in implied revenue loss.

The reconstructed revenue-loss measure should therefore be interpreted as a noisy proxy rather than an exact estimate of firm-specific revenue decline. In addition to the wage approximation, firms may also have differed in how accurately they reported expected revenue losses when applying for NOW support.

Despite this measurement error, the analysis focuses on aggregate patterns and relative firm rankings rather than exact firm-level revenue losses. Random over- and underestimation are therefore expected to offset each within sector aggregates removing the sector wage noise, therefore making average crisis exposure substantially more reliable than individual observations.

Appendix D: Horizon Choice

Choice of prediction horizons

The exit and employment-loss models use different prediction horizons because the underlying adjustment processes differ. This appendix explains the rationale for using a three-year horizon for the exit model and a one-year horizon for the employment-loss model.

Exit model: three-year horizon

The exit model uses a three-year outcome horizon for two reasons.

First, firms facing the same revenue shock are expected to exit at different times depending on their financial resilience. Some may fail after 12 months, while others survive for several years before exiting. A one-year horizon would therefore classify many firms that ultimately fail as unaffected by the shock, whereas a three-year horizon captures a broader range of crisis-induced exits.

Employment-loss model: one-year horizon

The employment-loss model uses a one-year outcome horizon because employment adjustment is expected to occur more rapidly than firm exit. Firms can reduce employment through layoffs, attrition or contract adjustments shortly after the onset of a shock. Extending the horizon would increasingly mix the immediate crisis response with subsequent recovery or longer-term structural changes, making it more difficult to isolate the employment adjustment associated with the initial crisis.

Appendix E: Why Round 1 Is Used for the Crisis-Induced Employment-Loss Analysis

This appendix explains why Round 1 NOW applications are used to estimate crisis-induced employment loss in Sections 5.1–5.3 rather than combining revenue losses across all eight subsidy rounds.

Why Round 1?

The framework estimates crisis-induced employment loss using firms' reported revenue losses. Round 1 revenue losses largely reflect the initial economic shock before the continued effects of the NOW programme could materially influence firms' financial position. In later rounds, however, reported revenue losses are no longer independent of the intervention itself. By supporting firms' liquidity and employment, NOW likely reduced subsequent revenue losses relative to the no-intervention scenario.

Because reported revenue loss is an input to the crisis component of the model, using later-round revenue losses would partially incorporate the effects of the programme into the estimated crisis shock, thereby understating the employment loss that might have occurred without support.

Implications

Round 1 therefore provides the closest available approximation to the initial untreated crisis shock and is used as the basis for estimating crisis-induced employment 5.1–5.3.

Using all eight NOW rounds remains a feasible alternative under the assumption of a longer crisis duration. However, doing so requires assumptions about the counterfactual evolution of firms' revenue losses after Round 1, since the observed revenue losses in later rounds are already affected by NOW support. The study therefore limits the counterfactual analysis

to the approximately three-month duration of Round 1, while using information from all eight rounds separately in Section 5.4 to evaluate alternative firm-ranking methods.

Direction of Bias

Both approaches are conservative. Restricting the analysis to Round 1 excludes additional crisis exposure that occurred later in the pandemic, while using observed revenue losses from later rounds incorporates the effects of the programme itself, reducing the measured crisis component. The true counterfactual crisis-induced employment loss in the absence of intervention is therefore likely to exceed both estimates.

Appendix F. Employment Loss Model: Specification and Validation

The employment loss model predicts the probability that a surviving firm reduces its headcount in the following year. This appendix explains the choice of outcome variable, the reasons for modelling the probability of shrinkage rather than the direct magnitude of employment loss, the sample restriction to firms with three or more employees, the estimation of the jobs-lost multiplier, and the validation of the resulting job loss estimates.

F.1 Why probability of shrinkage rather than magnitude

Employment loss among surviving firms is highly skewed. Most firms experience little or no employment reduction in a given year, while a small number account for large contractions. Directly modelling the magnitude of employment change therefore produces unstable predictions, including values that can fall below zero or exceed firm size.

A more fundamental problem emerged when revenue loss was added to the magnitude regression: the rev_loss coefficient showed the wrong sign, with larger revenue losses associated with less employment reduction rather than more. This may reflect survivorship bias. The regression sample includes only firms that survived, and firms surviving severe shocks may be more likely to recover employment in subsequent years, creating a positive relationship between shock severity and observed employment change.

In the logit specification predicting the probability of shrinkage, the revenue-loss coefficient is negative and correctly signed (−0.014), indicating that larger shocks increase employment risk as expected. Expected employment loss is then calculated as the predicted probability of shrinkage multiplied by the average number of jobs lost conditional on shrinkage. Revenue loss therefore affects aggregate employment loss through the probability term: larger revenue losses increase the probability of shrinkage and, consequently, the aggregate number of expected jobs lost. Table F3 summarises the trade-offs across the alternative approaches considered.

F.2 Sample restriction to firms with three or more employees

The training sample is restricted to firms with three or more employees. Firms with exactly two employees are excluded because a two-employee firm that reduces headcount falls to

one employee. Since firms with one employee are excluded from the panel throughout, this outcome is never observed in the data: the firm simply disappears from the employment register, making shrinkage indistinguishable from exit. Including two-employee firms as training observations would therefore allow the model to partially learn exit patterns through the shrinkage channel.

Firms with three employees can reduce headcount to two, which remains observable in the panel. In principle, a separate two-employee category could be modelled, but these firms represent a small share of employment and excluding them likely has a negligible effect on the aggregate results. At deployment, firms with two or fewer employees are assigned a shrinkage probability of zero, so their employment risk is captured exclusively through the exit channel.

F.3 Jobs lost multiplier

Expected jobs lost per shrinking firm is estimated directly from the training data as the average actual jobs lost among firms that shrank, grouped by exact firm size up to 19 employees. Above 19 employees, thin samples produce non-monotonic estimates, so a single pooled average is used for all firms with 20 or more employees. Table F1 reports the resulting multipliers.

jobs_group	avg_jobs_lost	Note
3	1.000	Minimum: can only lose 1 job
4	1.295	
5	1.449	
6	1.608	
7	1.743	
8	1.894	
9	2.025	
10	2.210	
11	2.275	
12	2.433	
13	2.500	
14	2.656	
15	2.809	
16	2.823	
17	2.974	
18	3.089	
19	3.190	Monotonic, individual rates used
20+	10.543	Non-monotonic above 19, pooled avg used

Table F1. *Average jobs lost per shrinking firm by firm size group. Individual rates are used for firms with 3–19 employees where the relationship is monotonically increasing. A pooled average of 10.54 is used for firms with 20 or more employees where thin samples produce non-monotonic estimates.*

F.4 Aggregate calibration

Table F2 validates the combined estimate (predicted shrinkage probability multiplied by the size-specific jobs lost multiplier) against actual observed job losses in the training data. Firms are ranked into deciles by predicted jobs lost and mean predicted and actual job loss are compared within each decile.

The ranking is monotonically correct across all ten deciles, confirming the model correctly orders firms by expected employment loss. At the aggregate level, total predicted jobs lost multiplied by 4.65 (the population average jobs lost per shrinking firm) equals 10.909, compared with an actual total of 10.816, a difference of less than 1 percent. This confirms near-perfect aggregate calibration for the training data. The multiplier approach is therefore used exclusively for aggregate expected job destruction estimates rather than firm-level point predictions.

Bucket	Mean pred	Mean actual	Pred × 4.65	Actual × 4.65
1	0.118	0.099	0.549	0.460
2	0.146	0.121	0.679	0.563
3	0.163	0.142	0.758	0.660
4	0.180	0.171	0.837	0.795
5	0.198	0.198	0.921	0.921
6	0.219	0.235	1.018	1.093
7	0.244	0.274	1.135	1.274
8	0.278	0.318	1.293	1.479
9	0.332	0.345	1.544	1.604
10	0.468	0.423	2.176	1.967
Total	**2.346**	**2.326**	**10.909**	**10.816**

Table F2. *Predicted vs actual shrinkage probability by decile, with implied jobs lost scaled by 4.65 (population average jobs lost per shrinking firm). Total predicted × 4.65 = 10.909 vs actual × 4.65 = 10.816, a difference of 0.9%, confirming near-perfect aggregate calibration at the population level.*

F.5 Summary of approaches considered

Table F3 summarises the trade-offs across the four approaches considered.

Approach	Benefit	Downside
OLS continuous regression	Direct magnitude prediction	Survivorship bias flips rev_loss sign; extreme residuals make fit poor
GLM probability only	Correct signs, AUC 0.65, stable coefficients	No job loss magnitude, needs second step
GLM × flat 4.65 avg	Aggregate ratio 1.003, simple	Biased when firm size distribution shifts between training and deployment

GLM × exact jobs_group multiplier (3–19 individual, 20+ pooled)	Aggregate ratio 1.003, size-corrected, monotonic up to 19 jobs	Non-monotonic above 20 jobs due to thin samples, pooled there

Table F3. *Employment loss magnitude approach comparison. The GLM probability model combined with a size-specific jobs lost multiplier is the preferred approach, producing an aggregate calibration ratio of 1.003 while correctly identifying the direction of crisis-induced shrinkage risk.*

Appendix G. Exit Model: Coefficients and Stability

This appendix reports the full exit-model coefficients and examines the stability of the estimated relationships under alternative crisis specifications. The stability analysis compares models with and without reported revenue loss while retaining the crisis dummy to evaluate whether both variables capture similar information and to justify the final model specification.

Variable	*Estimate*	*Std. Error*	*z value*	*Pr(>|z|)*	*Direction*
Firm characteristics					
(Intercept)	*-0.438*	*0.0078*	*-56.32*	*<2e-16 ****	—
log(number_of_jobs)	*-0.458*	*0.0011*	*-427.24*	*<2e-16 ****	↑ *Smaller firms more likely to exit*
age_bucket 0–3	*0.667*	*0.0035*	*191.27*	*<2e-16 ****	↑ *Young firms more likely to exit*
age_bucket 4–7	*0.253*	*0.0035*	*72.48*	*<2e-16 ****	↑
age_bucket 8–15	*0.065*	*0.0035*	*18.48*	*<2e-16 ****	↑
ft_share	*-0.353*	*0.0052*	*-67.32*	*<2e-16 ****	↓ *Higher full-time share reduces exit risk*
female_share	*-0.232*	*0.0072*	*-32.51*	*<2e-16 ****	↓ *Higher female share reduces exit risk*
female_ft_share	*0.093*	*0.0080*	*11.59*	*<2e-16 ****	↑
Crisis variable					
crisis_dummy	—	—	—	—	*Not included (collinear with rev_loss; see Table G2)*
rev_loss_market	*-0.016*	*0.0004*	*-36.53*	*<2e-16 ****	↑ *Larger revenue loss increases exit risk*
Sector fixed effects (ref: sector C)					
D -Energy supply	*0.857*	*0.0360*	*23.82*	*<2e-16 ****	↑
E -Water supply & waste management	*0.309*	*0.0187*	*16.56*	*<2e-16 ****	↑
F -Construction	*0.309*	*0.0044*	*70.69*	*<2e-16 ****	↑
G -Wholesale & retail trade	*0.327*	*0.0034*	*95.85*	*<2e-16 ****	↑
H -Transportation & storage	*0.499*	*0.0054*	*93.00*	*<2e-16 ****	↑
I -Accommodation & food service	*0.225*	*0.0044*	*50.83*	*<2e-16 ****	↑
J -Information & communication	*0.713*	*0.0052*	*137.67*	*<2e-16 ****	↑
K -Financial & insurance services	*0.691*	*0.0058*	*118.19*	*<2e-16 ****	↑

L -Real estate activities	*0.339*	*0.0075*	*45.05*	*<2e-16 ****	↑
M -Professional, scientific & technical	*0.516*	*0.0039*	*132.37*	*<2e-16 ****	↑
N -Administrative & support services	*0.583*	*0.0047*	*122.85*	*<2e-16 ****	↑
O -Public administration & defence	*0.755*	*0.0126*	*59.93*	*<2e-16 ****	↑
P -Education	*0.272*	*0.0059*	*46.18*	*<2e-16 ****	↑
Q -Human health & social work	*0.079*	*0.0047*	*16.83*	*<2e-16 ****	↑
R -Arts, recreation & entertainment	*0.283*	*0.0057*	*49.75*	*<2e-16 ****	↑
S -Other service activities	*0.226*	*0.0054*	*42.21*	*<2e-16 ****	↑
Province fixed effects (ref: Groningen)					
Friesland (Fryslân)	*-0.741*	*0.0062*	*-119.30*	*<2e-16 ****	↓ *Reduces exit risk vs reference province*
Drenthe	*-0.474*	*0.0064*	*-73.60*	*<2e-16 ****	↓
Overijssel	*-0.650*	*0.0055*	*-117.52*	*<2e-16 ****	↓
Flevoland	*-0.257*	*0.0070*	*-36.82*	*<2e-16 ****	↓
Gelderland	*-0.402*	*0.0049*	*-82.18*	*<2e-16 ****	↓
Utrecht	*-0.437*	*0.0053*	*-82.40*	*<2e-16 ****	↓
Noord-Holland	*-0.278*	*0.0046*	*-60.69*	*<2e-16 ****	↓
Zuid-Holland	*-0.566*	*0.0047*	*-120.52*	*<2e-16 ****	↓
Zeeland	*-0.503*	*0.0069*	*-73.28*	*<2e-16 ****	↓
Noord-Brabant	*-0.392*	*0.0047*	*-82.50*	*<2e-16 ****	↓
Limburg	*-0.627*	*0.0056*	*-112.57*	*<2e-16 ****	↓

Model fit: null deviance 8,888,136 on 8,531,861 df; residual deviance 8,369,337 on 8,531,826 df; AIC 8,369,409; n = 8,531,862.

Table G1. *Exit model GLM coefficients with direction of effect on firm exit probability. Sector fixed effects are relative to sector C -Manufacturing; all reported sectors increase exit risk relative to this baseline. Province fixed effects are relative to Groningen; all reported provinces reduce exit risk relative to the reference. The crisis dummy is excluded from the final exit model due to collinearity with revenue loss, and is retained only in the stability check (Table G2). Revenue loss enters as a negative number, so its negative coefficient implies that larger revenue losses increase exit risk. All reported coefficients significant at p < 0.001 (***). Reference category for age is 16+ years. Estimated on firms with 2 or more employees, training years 2000–2016.*

Coefficient Stability

Variable	Est. (base)	Est. (+ rev_loss)	% change	Sig. base	Sig. full
Firm characteristics					
(Intercept)	-0.395	-0.395	0.2%	*	*
log(number_of_jobs)	-0.457	-0.457	0.0%	*	*
age_bucket 0–3	0.677	0.676	-0.1%	*	*
age_bucket 4–7	0.238	0.237	-0.3%	*	*
age_bucket 8–15	0.065	0.065	0.0%	*	*

ft_share	-0.365	-0.365	0.0%	*	*
female_share	-0.228	-0.229	-0.1%	*	*
female_ft_share	0.088	0.089	0.2%	*	*
Crisis variables					
crisis_dummy	-0.212	-0.214	-0.8%	*	*
rev_loss_market	-	-0.005	-	-	*
Crisis dummy interactions					
age_bucket 0–3 × crisis	0.030	0.026	-14.4%	*	*
age_bucket 4–7 × crisis	0.225	0.219	-2.4%	*	*
age_bucket 8–15 × crisis	0.039	0.031	-20.7%	*	*
log(number_of_jobs) × crisis	0.003	0.003	-13.7%		
ft_share × crisis	0.067	0.067	-0.6%	*	*

Table G2. Stability of exit-model coefficients under alternative crisis specifications. *Est. (base)* reports the model including the crisis dummy and its interactions; *Est. (+ rev_loss)* additionally includes reported revenue loss. Structural firm-characteristic coefficients remain virtually unchanged after adding revenue loss, while the crisis interactions change only modestly and retain the same substantive interpretation. In contrast, the revenue-loss coefficient is small (-0.005) when the crisis dummy is included, suggesting that both variables capture similar crisis-period information. Although the crisis-dummy specification achieves a slightly lower training AIC (8,363,290 versus 8,369,409), the crisis dummy identifies the specific COVID-19 period and therefore cannot be used when extrapolating the model to future crises. The final exit model consequently omits the crisis dummy and retains reported revenue loss as the extrapolatable measure of crisis exposure. The log(number_of_jobs) × crisis interaction is not statistically significant; all other reported coefficients are significant at $p < .05$ (*). Sector and province fixed effects are omitted.

Appendix H Shrinkage Model Coefficients and Stability

This appendix reports the full employment-shrinkage model coefficients and examines the stability of the estimated relationships under alternative crisis specifications. The stability analysis compares models with and without reported revenue loss while retaining the crisis dummy to assess whether reported revenue loss contributes additional information beyond the common crisis effect and to justify the final model specification.

Variable	*Estimate*	*Std. Error*	*z value*	*Pr(>|z|)*	*Direction*
Firm characteristics					
(Intercept)	*-1.798*	*0.0136*	*-132.442*	*<2e-16 ****	*-*
log(number_of_jobs)	*0.437*	*0.0013*	*342.067*	*<2e-16 ****	*↑ Larger firms more likely to shrink*
age_bucket 0–3	*-0.120*	*0.0060*	*-19.808*	*<2e-16 ****	*↓ Young firms less likely to shrink*

age_bucket 4–7	*0.093*	*0.0054*	*17.114*	*<2e-16 ****	↑
age_bucket 8–15	*0.124*	*0.0054*	*23.096*	*<2e-16 ****	↑
ft_share	*-0.301*	*0.0100*	*-30.009*	*<2e-16 ****	↓ *Higher full-time share reduces shrink risk*
female_share	*0.111*	*0.0122*	*9.100*	*<2e-16 ****	↑ *Higher female share increases shrink risk*
female_ft_share	*-0.101*	*0.0137*	*-7.411*	*1.25e-13 ****	↓
Crisis variables					
crisis_dummy	*-0.053*	*0.0167*	*-3.202*	*0.001367 ***	↑ *Crisis increases shrink risk (negative coef on logit scale)*
rev_loss_market	*-0.014*	*0.0007*	*-19.637*	*<2e-16 ****	↑ *Larger revenue loss increases shrink risk*
Sector fixed effects (ref: sector C Manufacturing)					
D - Energy supply	*-0.268*	*0.0510*	*-5.248*	*1.54e-07 ****	↓
E - Water supply & waste management	*-0.060*	*0.0225*	*-2.647*	*0.008128 ***	↓
F - Construction	*0.124*	*0.0064*	*19.500*	*<2e-16 ****	↑
G - Wholesale & retail trade	*-0.059*	*0.0051*	*-11.608*	*<2e-16 ****	↓
H - Transportation & storage	*-0.088*	*0.0082*	*-10.715*	*<2e-16 ****	↓
I - Accommodation & food service	*-0.140*	*0.0067*	*-20.748*	*<2e-16 ****	↓
J - Information & communication	*-0.053*	*0.0100*	*-5.277*	*1.31e-07 ****	↓
K - Financial & insurance services	*0.079*	*0.0096*	*8.241*	*<2e-16 ****	↑
L - Real estate activities	*-0.015*	*0.0126*	*-1.156*	*0.247801*	↓ *(n.s.)*
M - Professional, scientific & technical	*0.027*	*0.0065*	*4.184*	*2.86e-05 ****	↑
N - Administrative & support services	*-0.075*	*0.0079*	*-9.499*	*<2e-16 ****	↓
O - Public administration & defence	*-0.429*	*0.0131*	*-32.714*	*<2e-16 ****	↓
P - Education	*-0.094*	*0.0074*	*-12.628*	*<2e-16 ****	↓
Q - Human health & social work	*-0.264*	*0.0066*	*-39.736*	*<2e-16 ****	↓
R - Arts, recreation & entertainment	*-0.067*	*0.0100*	*-6.714*	*1.90e-11 ****	↓
S - Other service activities	*0.030*	*0.0088*	*3.361*	*0.000778 ****	↑
Crisis dummy interactions					
sector D × crisis	*0.081*	*0.0957*	*0.850*	*0.395503*	↑ *(n.s.)*
sector E × crisis	*-0.167*	*0.0450*	*-3.702*	*0.000214 ****	↓
sector F × crisis	*-0.019*	*0.0129*	*-1.499*	*0.133951*	↓ *(n.s.)*
sector G × crisis	*-0.036*	*0.0096*	*-3.787*	*0.000152 ****	↓
sector H × crisis	*-0.057*	*0.0163*	*-3.476*	*0.000508 ****	↓
sector I × crisis	*-0.070*	*0.0130*	*-5.341*	*9.23e-08 ****	↓
sector J × crisis	*-0.009*	*0.0190*	*-0.455*	*0.648774*	↓ *(n.s.)*

sector K × crisis	*-0.102*	*0.0198*	*-5.148*	*2.63e-07 ****	↓
sector L × crisis	*0.120*	*0.0246*	*4.850*	*1.23e-06 ****	↑
sector M × crisis	*0.118*	*0.0122*	*9.661*	*<2e-16 ****	↑
sector N × crisis	*0.058*	*0.0150*	*3.846*	*0.000120 ****	↑
sector O × crisis	*0.061*	*0.0264*	*2.325*	*0.020086 **	↑
sector P × crisis	*0.101*	*0.0139*	*7.282*	*3.30e-13 ****	↑
sector Q × crisis	*0.082*	*0.0117*	*6.986*	*2.82e-12 ****	↑
sector R × crisis	*0.075*	*0.0189*	*3.942*	*8.07e-05 ****	↑
sector S × crisis	*-0.013*	*0.0169*	*-0.749*	*0.454104*	↓ *(n.s.)*
age_bucket 0–3 × crisis	*-0.187*	*0.0097*	*-19.255*	*<2e-16 ****	↓ *Young firms face less additional shrink risk in crisis*
age_bucket 4–7 × crisis	*-0.322*	*0.0098*	*-32.731*	*<2e-16 ****	↓
age_bucket 8–15 × crisis	*-0.238*	*0.0080*	*-29.779*	*<2e-16 ****	↓
ft_share × crisis	*0.218*	*0.0121*	*18.086*	*<2e-16 ****	↑ *Higher full-time share increases shrink risk in crisis*
log(number_of_jobs) × crisis	*0.060*	*0.0025*	*23.793*	*<2e-16 ****	↑ *Larger firms face more shrink risk in crisis*
Province fixed effects (ref: Groningen)					
Friesland (Fryslân)	*-0.166*	*0.0089*	*-18.531*	*<2e-16 ****	↓
Drenthe	*-0.220*	*0.0096*	*-23.019*	*<2e-16 ****	↓
Overijssel	*-0.050*	*0.0080*	*-6.287*	*3.23e-10 ****	↓
Flevoland	*0.206*	*0.0101*	*20.420*	*<2e-16 ****	↑
Gelderland	*0.111*	*0.0074*	*14.903*	*<2e-16 ****	↑
Utrecht	*-0.085*	*0.0079*	*-10.797*	*<2e-16 ****	↓
Noord-Holland	*-0.495*	*0.0075*	*-66.081*	*<2e-16 ****	↓
Zuid-Holland	*-0.211*	*0.0073*	*-29.042*	*<2e-16 ****	↓
Zeeland	*-0.163*	*0.0099*	*-16.506*	*<2e-16 ****	↓
Noord-Brabant	*-0.238*	*0.0074*	*-32.226*	*<2e-16 ****	↓
Limburg	*-0.209*	*0.0081*	*-25.911*	*<2e-16 ****	↓

Model fit: null deviance 4,769,755 on 4,376,663 df; residual deviance 4,530,503 on 4,376,606 df; AIC 4,530,619; n = 4,376,664.

Table H1. *Employment loss model GLM coefficients with direction of effect on shrinkage probability. Sector fixed effects are relative to sector C - Manufacturing; province fixed effects are relative to Groningen. Crisis dummy interactions allow the crisis effect to vary by sector, firm age, size, and full-time share. Revenue loss enters as a negative number, so the negative coefficient (−0.014) implies larger revenue losses increase shrinkage risk. Significance codes: *** p<0.001, ** p<0.01, * p<0.05; coefficients marked (n.s.) are not significant at p<0.05.*

Coefficient Stability

Variable	Est. (base)	Est. (+ rev_loss)	% change	Sig. base	Sig. full
Firm characteristics					
(Intercept)	-2.627	-2.625	0.1%	*	*

log(number_of_jobs)	0.673	0.672	0.0%	*	*
age_bucket 0–3	-0.188	-0.190	-1.1%	*	*
age_bucket 4–7	0.079	0.076	-3.2%	*	*
age_bucket 8–15	0.126	0.126	-0.1%	*	*
ft_share	-0.251	-0.252	-0.2%	*	*
female_share	0.214	0.213	-0.4%	*	*
female_ft_share	-0.107	-0.106	0.8%	*	*
Crisis variables					
crisis_dummy	-0.055	-0.070	-26.2%	*	*
rev_loss_market	-	-0.014	-	-	*
Crisis dummy interactions					
age_bucket 0–3 × crisis	-0.209	-0.220	-5.3%	*	*
age_bucket 4–7 × crisis	-0.330	-0.347	-5.2%	*	*
age_bucket 8–15 × crisis	-0.219	-0.246	-12.2%	*	*
ft_share × crisis	0.255	0.253	-0.9%	*	*
log(number_of_jobs) × crisis	0.060	0.059	-1.0%	*	*

Table H2. *Stability of employment-loss model coefficients under alternative crisis specifications. Est. (base) reports the model including the crisis dummy and its interactions; Est. (+ rev_loss) additionally includes firm-declared revenue loss. Structural coefficients remain virtually unchanged after adding revenue loss, while the revenue-loss coefficient remains substantial and statistically significant. This indicates that reported revenue loss and the crisis dummy capture complementary information rather than the same crisis signal. Consequently, both variables are retained in the final employment-loss model. Sector and province fixed effects are omitted.*

Appendix J

Sector	Description	Wage W_comp (€)	Cap φ	Duration Du (mo.)	Wage measured 2nd digit (codes)	Duration measured 1st digit (codes)
A	Agriculture, forestry & fishing	3,764	0.99	9.2	3/3	Yes
B	Mining & quarrying	10,198	0.48	12.2	0/3	No
C	Manufacturing	5,184	0.91	8.4	24/24	24/24

D	Energy supply	6,857	0.71	12.2	1/1	No
E	Water & waste	5,350	0.92	8.4	4/4	No
F	Constructio n	5,167	0.92	10.2	3/3	Yes
G	Wholesale & retail	3,362	1.00	9.0	3/3	Yes
H	Transport & storage	4,088	0.99	7.7	5/5	Yes
I	Hospitality	1,283	1.00	7.5	0/2	2/2
J	Information & communicat ion	5,525	0.88	12.2	6/6	No
K	Financial services	7,777	0.63	12.2	3/3	Yes
L	Real estate	4,855	1.00	9.1	1/1	No
M	Professional services	4,812	0.92	9.1	7/7	Yes
N	Administrati ve services	3,043	1.00	7.1	6/6	Yes
O	Public administrati on	5,309	0.92	10.4	1/1	Yes
P	Education	4,364	1.00	8.7	1/1	Yes
Q	Health & social work	3,268	1.00	9.1	3/3	Yes
R	Arts & recreation	2,819	1.00	7.7	4/4	Yes
S	Other services	3,183	1.00	9.1	3/3	Yes

Table J1. *Sector-level inputs to the cost calculation (2019-2020). W_comp is the average monthly wage including employer social contributions; φ is the benefit-cap factor (φ < 1 where the sector wage exceeds the maximum dagloon, so benefits apply only up to the ceiling); Du is the expected unemployment duration in months. The final two columns report how many of the sector's constituent two-digit SBI codes use a directly measured value rather than one borrowed from a parent or related sector (shaded where any are borrowed). Wages from CBS National Accounts; durations from UWV records. Values are shown at sector-letter level as unweighted means across two-digit codes; the cost calculation applies the underlying two-digit values, which are provided in the replication data.*

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